Cancer as Escalation Chain Severance Under Bounded Context

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Abstract

We apply the bounded-context framework of Papers 1–4 to multicellular organization, proposing that cancer is a predictable failure mode of any multi-level escalation hierarchy. In a multicellular organism, individual cells participate in an escalation chain – cell → tissue → organ → organism – where each level extends the effective bounded context C_n^{eff} of the level below via intercellular signaling. We argue that severing the chain at level k forces the cell back to $C_n^{(k)}$-level optimization, where the action space available under cell-level bounded context converges on the recognized hallmarks of cancer: aerobic glycolysis, unconstrained proliferation, immune evasion, and motility. This convergence is not atavistic reversion to ancient programs but *structural necessity*. The same constraint ceiling that governed unicellular life now governs the severed cell. The framework offers structural explanations for three phenomena that existing theories address incompletely: (1) the velocity of malignant transformation (structural failure, not stochastic accumulation), (2) the convergent phenotype across tissue types (same constraint, same necessary structure), and (3) the therapeutic resistance of established tumors (locally rational optimization under cell-level C_n). Clinical oncology is already trending toward the therapeutic strategies this framework predicts – differentiation therapy, immunotherapy, and microenvironment normalization – but largely without a unified theoretical justification. We propose that the bounded-context framework provides one, and offer testable predictions that distinguish this account from the somatic mutation theory, the atavistic theory, and the tissue organization field theory.

1 Introduction

Cancer remains the central unsolved problem of multicellular biology. Three major theoretical frameworks compete to explain it, each capturing part of the phenomenon while missing the rest.

The *somatic mutation theory* (SMT) holds that cancer arises from accumulated genetic mutations that progressively disable growth controls [17, 14]. SMT explains the stochastic component of cancer incidence and the role of specific oncogenes and tumor suppressors. It does not explain the *velocity* of malignant transformation: why cancers from different tissues, with different mutational profiles, converge on the same phenotypic hallmarks [7] far faster than random mutation would predict.

The *atavistic theory* holds that cancer cells reactivate ancient genetic programs from the unicellular era, effectively reverting to a pre-multicellular phenotype [4, 16]. The phylostratigraphic evidence is striking: genes upregulated in cancer are disproportionately ancient, dating to unicellular ancestors [16]. But the theory lacks a mechanism: *why* would a modern cell access billion-year-old programs, and *how* are those programs still coherent after extensive evolutionary modification?

The *tissue organization field theory* (TOFT) holds that cancer is not a disease of individual cells but of tissue organization: the breakdown of intercellular signaling and tissue architecture enables

default proliferative behavior [15]. TOFT correctly identifies the organizational level at which cancer operates but lacks a formal account of *why* organizational breakdown produces the specific phenotypes it does.

This paper proposes that all three theories describe aspects of a single structural phenomenon: **escalation chain severance**. The bounded-context framework of Papers 1–4 [8, 9, 10, 11] provides a formal lens through which to unify them. We model multicellular organization as a multi-level escalation hierarchy where each level extends the effective bounded context of the level below. Severing the chain is a structural failure that forces cell-level optimization. The convergent phenotype, on this account, is not inherited from ancestors but *necessary* under cell-level bounded context. The same constraint that shaped unicellular life now shapes the severed cell.

Scope and epistemology. This paper is a theoretical application of the bounded-context framework to cancer biology. Papers 1–4 derive their results from formal proofs; the present paper applies that framework to a biological domain where the mapping from formalism to biology is conjectural. We do not claim to prove anything about cancer. We claim that the framework generates a coherent structural account that (a) unifies three competing theories, (b) aligns with the direction clinical oncology is already moving, and (c) produces testable predictions that distinguish it from existing accounts. The value of the paper is the structural reasoning and the predictions, not empirical validation, which is the province of cancer biologists and clinicians.

2 Formal Framework

We extend the entity framework of Papers 1–4 to multi-level biological organization.

Definition 1 (Biological Entity at Level k)**.** A biological entity at organizational level k is $E_k = (K_k, A_k, \sigma_k, \gamma_k)$ with bounded active context $C_n^{(k)}$, where:

- K_k is the knowledge base at level k (the regulatory state accessible for inference: gene expression profiles, signaling molecule concentrations, tissue-level morphogen gradients).
- A_k is the action space at level k (the executable responses — protein synthesis, cell division, apoptosis, migration, differentiation signals).
- σ_k is the sensing function at level k (receptors, gap junctions, paracrine signaling).
- γ_k is the agency at level k (the probability that E_k engages its adaptive mechanism in response to signals).

The organizational levels in a metazoan are:

$$\begin{aligned} k &= 0 : \text{cell} \\ k &= 1 : \text{tissue} \\ k &= 2 : \text{organ} \\ k &= 3 : \text{organism} \end{aligned}$$

Definition 2 (Escalation Chain)**.** An *escalation chain* is a sequence of coupled entities $E_0, E_1, \ldots, E_m$ where level $k+1$ receives escalation signals from level k when E_k encounters states exceeding its local capacity:

$$U_k(w, K_k) > U_{\text{threshold}}^{(k)} \;\Rightarrow\; \text{escalate to } E_{k+1}$$

This is the biological instantiation of escalation dominance [9]: higher levels engage only when lower levels fail, achieving $O(1)$ steady-state context cost.

Definition 3 (Effective Context Extension)**.** When E_k is coupled to E_{k+1} via an intact escalation chain, the *effective bounded context* of E_k is:

$$C_n^{\text{eff}}(E_k) = C_n^{(k)} + \sum_{j=k+1}^{m} \lambda_j \cdot C_n^{(j)}$$

where $\lambda_j \in [0, 1]$ is the *aggregate* coupling strength between levels, summarizing the fidelity and bandwidth of intercellular signaling across all channels at level j (Section 2.1 decomposes λ_j into per-channel components). $\lambda_j = 1$ represents perfect coupling (all of E_j's context is available to E_k via signaling); $\lambda_j = 0$ represents complete decoupling.

The key property: $C_n^{\text{eff}} > C_n^{(k)}$. The cell "borrows" context from higher organizational levels. A liver cell does not need to independently represent the organism's metabolic state; it receives that information via hormonal and neural signaling from the organ and organism levels.

Definition 4 (Chain Severance)**.** *Chain severance at level k* occurs when the coupling $\lambda_{k+1} \to 0$: the entity E_k loses its connection to all higher organizational levels. Post-severance:

$$C_n^{\text{eff}}(E_k) = C_n^{(k)}$$

The cell's effective context collapses to its intrinsic cell-level bound.

2.1 Biological Instantiation of the Coupling Coefficients

The abstract coupling coefficients λ_j correspond to specific, measurable intercellular communication mechanisms at each level of the hierarchy.

Level $0 \to 1$ (cell $\to$ tissue coupling). This is the primary coupling layer and the level at which most cancers sever. The coupling mechanisms are:

- **Gap junctions** (connexins, especially Cx43): direct cytoplasmic channels between adjacent cells that share ions, metabolites, and second messengers. Gap junctional intercellular communication (GJIC) is the highest-bandwidth cell-tissue coupling channel. Connexin-43 downregulation is among the earliest events in many epithelial cancers [1].
- **E-cadherin** (CDH1): homophilic adhesion molecule that mediates contact inhibition of proliferation. Loss of E-cadherin is the defining event of epithelial-mesenchymal transition (EMT) and directly severs the contact-dependent component of λ_1 [7].
- **Juxtacrine signaling** (Notch–Delta): direct cell-cell contact signaling that maintains tissue identity and differentiation state. Requires physical adjacency, destroyed by loss of tissue architecture.
- **Integrin–ECM attachment**: anchorage to the extracellular matrix and basement membrane. Loss of anchorage dependence (anoikis resistance) severs the structural component of tissue coupling.

Level $1 \to 2$ (tissue $\to$ organ coupling).

- **Paracrine signaling**: local growth factors and morphogen gradients (Wnt, Hedgehog, BMP, TGF-β) that coordinate tissue-level organization within the organ context.
- **Extracellular matrix architecture**: the structural scaffold that encodes tissue-level spatial organization and transmits mechanical signals.
- **Local vasculature**: nutrient and oxygen delivery that ties tissue metabolism to organ-level regulation. Vascular normalization restores this coupling channel.

Level $2 \to 3$ (organ $\to$ organism coupling).

- **Endocrine signaling**: systemic hormones (insulin, estrogen, cortisol, thyroid hormone) that couple organ function to organismal state. Hormone-dependent cancers (breast, prostate) retain partial λ_3 coupling, explaining their relative treatability.
- **Immune surveillance**: T-cells, NK cells, and macrophages that enforce organism-level quality control on aberrant cells. Immune checkpoint molecules (PD-L1, CTLA-4) are the molecular implementation of this coupling channel.
- **Neural innervation**: autonomic nervous system regulation of organ function and tissue homeostasis.
- **Circadian regulation**: organism-level temporal coordination via the SCN-driven hormonal cascade.

More precisely, each aggregate λ_j decomposes into per-channel components. Chain severance can be partial: a cell may lose gap junctional coupling ($\lambda_1^{\text{GJIC}} \to 0$) while retaining cadherin-mediated contact inhibition ($\lambda_1^{\text{CDH1}} > 0$). This partial severance produces the intermediate phenotypes observed in pre-malignant lesions. Complete severance across all channels at a given level produces the full malignant phenotype.

3 Main Result

Proposition 1 (Escalation Chain Severance Produces Convergent Malignancy)**.** *Let E_0 be a cell in a multicellular organism with intact escalation chain $E_0 \to E_1 \to \cdots \to E_m$, modeled as an entity under the bounded-context framework. If chain severance occurs at level $k = 0$ (cell loses coupling to tissue), then:*

Part I. *E_0 optimizes under $C_n^{(0)}$ alone. By graph necessity [8], it maintains only the propositions serviceable within cell-level bounded context.*

Part II. *By K/A inseparability [10], the knowledge base K_0 contracts to index only cell-level actions A_0: those executable without tissue-level coordination.*

Part III. *By the escalation convergence result [9], with no higher level to escalate to, all optimization pressure concentrates at E_0. The cell becomes the terminal level of its own hierarchy.*

Part IV. *The action space under cell-level $C_n^{(0)}$ converges across cell types. $C_n^{(0)}$ is small relative to the full repertoire of cell-level actions, so the asymmetric retention criterion of [10] forces displacement toward the same survival-essential set: with limited slots and the same survival-critical actions competing for them, tissue-specific content is progressively displaced by actions with nonzero expected benefit under cell-level optimization.*

Argument.

Part I: Context Collapse.

Before severance, E_0 operates with effective context $C_n^{\text{eff}} = C_n^{(0)} + \sum_{j=1}^{m} \lambda_j C_n^{(j)}$. The cell maintains propositions that reference tissue-level and organism-level states: differentiation signals, growth factor dependencies, contact inhibition cues, apoptotic triggers from the tissue microenvironment. These propositions are serviceable *because* the escalation chain provides the context in which they have grounding.

After severance ($\lambda_j = 0$ for $j \geq 1$), these propositions lose their grounding: they reference states that E_0 can no longer observe (tissue integrity, organ function, organismal need). By graph necessity [8], propositions without grounding in observable states cannot be maintained under bounded context without displacing grounded propositions. The retention criterion of [10] – retain when benefit is uncertain, prune when benefit is certainly zero – applies: the benefit of tissue-referencing propositions is certainly zero to a cell that cannot observe tissue state. They are pruned.

K_0 contracts to propositions grounded in cell-level observables: local nutrient concentration, immediate mechanical environment, intracellular state.

Part II: Action Space Contraction.

By K/A inseparability [10], K_0 indexes A_0 and vice versa. As K_0 contracts to cell-level propositions, A_0 contracts to cell-level actions. Actions that require tissue-level coordination – responding to contact inhibition, executing apoptosis on organismal command, maintaining differentiated quiescence – lose their knowledge support. They are not "disabled" by mutation; they become *inaccessible* because the knowledge required to execute them (tissue state, organismal need) is no longer in K_0.

What remains in A_0 are actions executable with cell-level knowledge alone:

- **Proliferation**: cell division requires only local nutrient availability and intracellular state; no tissue-level coordination needed.
- **Metabolic autonomy**: aerobic glycolysis (the Warburg effect [18]) is the metabolic mode with the fewest tissue-level *dependencies*: it does not require reliable tissue-level oxygen delivery infrastructure. Critically, cancer cells use glycolysis even when oxygen is present—not because oxygen is unavailable, but because after severance the cell lacks tissue-level information about oxygen supply reliability and defaults to the metabolic strategy that requires no such information. This does not imply mitochondrial shutdown (many cancer cells retain functional oxidative phosphorylation and use both pathways), but glycolytic capacity provides metabolic independence from tissue-level coordination.
- **Motility**: movement in response to local nutrient gradients requires only cell-level sensing.
- **Immune evasion**: both passive and active mechanisms. MHC-I surface markers, maintained by tissue-level signaling, decay when that signaling is lost. Additionally, cancer cells actively upregulate PD-L1 in response to immune attack (IFN-γ signaling)—an adaptive response that requires only cell-level sensing of immune pressure, not tissue-level coordination.

Part III: Terminal Optimization.

In the intact organism, the cell's optimization is bounded by escalation: problems exceeding $C_n^{(0)}$ escalate to tissue, organ, or organism level. The cell does not need to solve every problem; it only needs to handle its local scope and signal failure upward.

After severance, there is no higher level. By the escalation convergence result [9], this means all optimization pressure concentrates at E_0. The cell is no longer a component in a hierarchy; it is a complete autonomous entity. Its optimization target is its own persistence, gradient descent on U at cell level [11], with no constraint from above.

This is precisely the optimization regime of a unicellular organism: an autonomous entity with cell-level C_n, optimizing for its own survival under bounded context. The cell has not "reverted" to a unicellular program. It has been *forced into the same optimization regime* by the loss of its escalation chain.

Part IV: Convergence.

The convergence argument is not that all severed cells must have identical knowledge graphs. Graph necessity [8] requires graph *structure*, not identical graph *content*. The convergence is at the level of the action space, and it follows from the interaction of bounded context with the asymmetric retention criterion.

$C_n^{(0)}$ is small relative to the full repertoire of possible cell-level actions. Under the displacement criterion of [10], when K_0 is at capacity, a proposition p is displaced by q only when $\mathbb{E}[G(a_q, K)] > \mathbb{E}[G(a_p, K)]$, strictly higher marginal survival value. After severance, tissue-specific propositions have zero marginal value (they reference unobservable states), while cell-level survival actions (proliferation, nutrient acquisition, damage repair) have nonzero value regardless of tissue of origin. The retention criterion therefore drives displacement in the same direction for all severed cells: tissue-specific content is replaced by cell-level survival content.

The convergence is not perfect; residual tissue-specific K_0 content produces tissue-specific variation, especially early in progression. But the attractor is the same: the set of actions with highest marginal survival value under $C_n^{(0)}$. This set is constrained by cell biology, not by tissue of origin, so it converges across cancer types.

This offers a structural account for why Hanahan and Weinberg's hallmarks [7] are observed across cancer types: they are not shared mutations or shared ancient programs, but the *convergent action space* that the retention criterion selects under cell-level bounded context.

Status of this result. Proposition 1 is a conceptual argument, not a formal derivation. Each Part invokes a result from the formal chain (Papers 1–4) and applies it to the biological setting by analogy. The analogies are specific and testable—Section 8 states falsifiable predictions—but the Proposition does not prove that the hallmarks are the *unique* convergent action set under cell-level C_n. It argues that convergence to a small, survival-essential set is structurally predicted, and that the observed hallmarks are consistent with that prediction.

4 Corollaries

Corollary 2 (Velocity of Malignant Transformation)**.** *Malignant transformation does not require the accumulation of a specific set of mutations. It requires chain severance: the loss of coupling to the tissue-level escalation chain. Chain severance can occur through a single event (loss of a key signaling receptor, disruption of gap junctions, escape from the tissue microenvironment) rather than through the sequential accumulation of independent mutations.*

Argument. The somatic mutation theory requires multiple independent hits [17]: each hallmark capability must be separately acquired through mutation. Under our framework, the hallmarks are not independently acquired; they are jointly *released* when the escalation chain severs. A

single event that decouples the cell from tissue-level signaling simultaneously enables all hallmark behaviors, because all of them are cell-level actions that were *suppressed* by tissue-level coordination, not *absent* from the cell's intrinsic repertoire.

This explains why cancer can progress rapidly from a single initiating event: the "multi-hit" appearance is a consequence of progressive chain severance (each hit weakens coupling further) rather than independent capability acquisition. The velocity is structural, not stochastic.

Compatibility with multi-hit observations. The framework does not contradict the observation that multiple mutations are typically found in cancers. Rather, it reinterprets their role: mutations are mechanisms of chain severance (disrupting signaling pathways, disabling apoptotic responses to tissue-level commands), not independent capability acquisitions. The "multi-hit" pattern reflects progressive decoupling: each mutation weakens a different coupling channel λ_j, rather than independent construction of each hallmark.

Corollary 3 (Therapeutic Resistance)**.** *Established tumors exhibit persistent therapeutic resistance because the severed cell is* locally rational*: it is optimizing correctly under its own bounded context.*

Argument. Therapies that target specific mutations (targeted therapy) or specific proliferative mechanisms (chemotherapy) impose selection pressure on an entity that is already performing gradient descent on U at cell level [11]. The cell's $\gamma > 0$ (Paper 4, [11]) guarantees continued engagement with adaptive mechanisms. Its learning efficiency $\eta_M > 0$ (the rate of improvement per unit of learning signal; [11]) guarantees learning from therapeutic pressure. The therapy becomes a new environmental signal that the cell incorporates into its optimization.

This explains why resistance is the norm rather than the exception: the cell is not a broken machine to be fixed but an autonomous entity optimizing under its own constraints. Therapy that does not restore the escalation chain merely changes the optimization landscape without changing the optimization regime.

Effective therapy, on this account, must either:

1. **Restore the chain**: re-establish tissue-level coupling so that $C_n^{\text{eff}} > C_n^{(0)}$, returning tissue-level actions to the cell's accessible action space. Differentiation therapy (e.g., all-trans retinoic acid in acute promyelocytic leukemia) operates this way.

2. **Exploit the constraint**: target cell-level actions that are *necessary* under $C_n^{(0)}$, not just contingently adopted. If an action is necessary under cell-level bounded context, the cell cannot evolve away from it without leaving the $C_n^{(0)}$ optimization regime.

Corollary 4 (The Atavistic Pattern Without Atavistic Causation)**.** *The phylostratigraphic signal – cancer genes being disproportionately ancient [16] – is real but does not imply atavistic reversion. It reflects convergent necessity.*

Argument. Genes that are ancient are precisely the genes that were optimized for cell-level bounded context during the unicellular era. When a modern cell undergoes chain severance and is forced back to cell-level optimization, the *same genes* become relevant again, not because an ancient program is "reactivated" but because the same constraint ($C_n^{(0)}$) selects for the same solutions.

The atavistic theory [4] correctly identifies the phenotypic pattern (cancer cells resemble unicellular organisms) and the genetic evidence (ancient genes are upregulated). Its error is causal: it infers

historical reversion where the mechanism is structural convergence. The genes are not activated *because* they are ancient; they are activated because they encode the actions that are necessary under cell-level bounded context, and they happen to be ancient because unicellular organisms faced the same constraint.

This distinction is testable: the atavistic theory predicts that the ancient gene set forms a coherent "unicellular program" that is activated as a unit. Our framework predicts that the set of genes activated in any given cancer will be determined by the specific $C_n^{(0)}$-level optimization landscape, and will include ancient genes *to the extent that* they encode cell-level actions, but will also include recent genes that happen to serve cell-level optimization (e.g., genes co-opted for immune evasion that evolved in the multicellular era).

Corollary 5 (Metastasis as Optimal Foraging Under Cell-Level C_n)**.** *Metastasis is not an acquired "capability" but the default exploratory behavior of an autonomous entity optimizing under bounded context in a spatially heterogeneous environment.*

Argument. In an intact organism, cell motility is constrained by tissue-level signaling: basement membrane attachment, contact inhibition, tissue boundary maintenance. These constraints are maintained by the escalation chain. The tissue level enforces spatial organization that individual cells could not compute independently.

After chain severance, the cell has no spatial constraint beyond its local environment. By the non-termination result of [11] (Theorem 1, Part II), the reachable trajectory space $T_{\text{reachable}}(K_0)$ grows monotonically with K_0. A cell in a nutrient-poor local environment will, through gradient descent on U, discover that movement to nutrient-rich regions reduces uncertainty about survival. Metastasis is exploration, the expansion of $T_{\text{reachable}}$ into new spatial regions, driven by the same gradient that drives all learning [11].

The "invasion-metastasis cascade" of epithelial-mesenchymal transition, intravasation, survival in circulation, and colonization at distant sites is not a sequence of independently acquired capabilities. It is the progressive unfolding of cell-level exploration in a spatially structured environment, where each step reveals new states that reduce local uncertainty.

Corollary 6 (Transplant Rejection as Chain Mismatch)**.** *Organ transplant rejection is not chain severance but* chain mismatch*: the transplanted tissue has intact internal coupling (levels* $0 \to 1$*) but mismatched organism-level coupling (level* $2 \to 3$*). The immune system's rejection response is the escalation chain functioning correctly, detecting cells whose coupling markers do not match the recipient organism.*

Argument. A transplanted organ retains its internal tissue organization: the cells are coupled to each other and to tissue-level architecture via gap junctions, cadherins, and paracrine signaling. The λ_1 coupling channels are intact. What is mismatched is $\lambda_3^{\text{immune}}$: the organism-level immune surveillance system recognizes the transplanted cells' MHC markers as foreign, not coupled to *this* organism's escalation chain.

The organism cannot distinguish between chain severance (cancer) and chain mismatch (transplant). In both cases, the level $2 \to 3$ coupling signal is absent or aberrant, and the immune enforcement mechanism responds identically: elimination.

This reframes the immunosuppression tradeoff structurally. Immunosuppressive therapy (cyclosporine, tacrolimus) weakens the organism-level coupling enforcement $\lambda_3^{\text{immune}} \to 0$ globally. This permits

the mismatched organ but simultaneously disables the surveillance mechanism that detects genuine chain severance. The well-documented 2–5× elevation in cancer incidence among transplant recipients [3] is the predicted consequence: weakened chain enforcement permits both the intended mismatch *and* unintended severance events.

Tolerance induction – protocols that establish mixed hematopoietic chimerism or regulatory T-cell-mediated acceptance – operates differently. Rather than weakening enforcement, it re-establishes matching coupling: the recipient's immune system learns to recognize the transplanted tissue as part of its own escalation chain. The coupling $\lambda_3^{\text{immune}}$ is restored for the transplant specifically, while global enforcement remains intact. The framework predicts that tolerance induction should produce better long-term outcomes than chronic immunosuppression, not only for graft survival but for overall survival, because it preserves the organism-level chain enforcement that prevents cancer. Early clinical evidence from mixed chimerism protocols is consistent with this prediction [5].

HLA matching reduces the mismatch at the coupling interface itself, minimizing the signal that triggers enforcement without weakening enforcement. This is why HLA-matched transplants require less immunosuppression and have better outcomes: less mismatch means less enforcement to suppress.

5 Mechanisms of Chain Severance

Chain severance is not a single event but a spectrum of decoupling mechanisms, each reducing λ_j toward zero:

Genetic mechanisms. Mutations in signaling receptors (e.g., E-cadherin loss), tumor suppressor pathways (e.g., p53, Rb), or growth factor dependencies reduce the cell's capacity to receive tissue-level signals. These are the mechanisms identified by SMT [17]. In our framework, they are not causes of cancer but *mechanisms of chain severance*: each mutation weakens a specific coupling channel.

Microenvironmental mechanisms. Chronic inflammation, fibrosis, or tissue damage disrupts the signaling infrastructure through which higher levels communicate with cells. This is the mechanism identified by TOFT [15]: the tissue architecture that maintains coupling is destroyed, and cells decouple even without intrinsic mutations. The coupling $\lambda_j \to 0$ occurs not because the cell's receivers are broken but because the signal is no longer transmitted.

Epigenetic mechanisms. Methylation changes, chromatin remodeling, or transcriptional reprogramming can silence the genes encoding tissue-level signaling components without genomic mutation. This produces chain severance through altered gene expression rather than altered sequence.

Spatial mechanisms. Physical separation from the tissue microenvironment – through basement membrane breach, for instance – removes the cell from the signaling field. The cell's coupling hardware is intact; the coupling medium is absent.

All four mechanisms produce the same outcome: $\lambda_j \to 0$, effective context collapse to $C_n^{(0)}$, and optimization under cell-level bounded context. The convergent phenotype (the hallmarks) follows regardless of *which* mechanism severed the chain. This explains why cancers arising from different mechanisms (hereditary, environmental, sporadic) converge on the same phenotypic endpoints.

6 The Encoding Hierarchy in Cancer

The encoding hierarchy of [11] has a specific consequence for cancer that connects to the empirical phenomenon of tumor heterogeneity.

In the intact organism, actions at each encoding level have been validated to the threshold $\theta_i = 1 - \varepsilon/C_i$. The genomic level ($L_0$) encodes actions validated across evolutionary timescales. The epigenetic level (L_1) encodes tissue-specific programs validated across developmental timescales. The active context level (L_n) encodes moment-to-moment responses.

After chain severance, the encoding hierarchy is disrupted at all levels above the cell:

- **Organism-level encodings** (hormonal responses, circadian regulation, immune cooperation) lose their validation context. They were validated under C_n^{eff}; they are now evaluated under $C_n^{(0)}$. Many fail the retention criterion.
- **Tissue-level encodings** (differentiation state, contact inhibition, growth factor dependence) similarly lose validation context.
- **Cell-level encodings** (basic metabolism, division machinery, DNA repair) retain their validation. They were validated under $C_n^{(0)}$ originally and remain valid.

The result is *selective devalidation*: the encoding hierarchy does not collapse uniformly but selectively loses its higher-level content. What remains is a cell operating on cell-level encodings, the same encodings that a unicellular organism would maintain. This is the encoding-level explanation for the atavistic gene expression pattern observed by Trigos et al. [16]: ancient genes are preferentially retained not because they are "reactivated" but because they are the ones whose validation survives context collapse.

Tumor heterogeneity. The progressive nature of context collapse explains intratumor heterogeneity. Cells at different stages of chain severance retain different amounts of higher-level encoding. Early-stage tumor cells may retain partial tissue-level coupling (some $\lambda_j > 0$), producing a mixed phenotype. As severance progresses, the phenotype converges toward the cell-level optimum. This gradient of severance within a single tumor produces the phenotypic heterogeneity observed empirically, without requiring different clonal origins for different phenotypes.

Clonal evolution [14] describes the *dynamics* of this process: cells with more complete severance have higher fitness in the decoupled regime and are selected. But the direction of evolution is not random; it converges on the cell-level optimum, which is structurally determined by $C_n^{(0)}$.

7 Engagement with Prior Work

Hanahan and Weinberg [6, 7]: Hallmarks of Cancer. The hallmarks – sustaining proliferative signaling, evading growth suppressors, resisting cell death, enabling replicative immortality, inducing angiogenesis, activating invasion and metastasis, plus the enabling characteristics and emerging hallmarks of the 2011 update – are described as "capabilities" that must be "acquired." Our framework reinterprets them: they are not acquired but *revealed*. They are the necessary action space of any entity optimizing under cell-level bounded context. The hallmarks are universal across cancer types because $C_n^{(0)}$ is universal across cell types. What differs across cancers is the *route* of chain severance (which coupling channels fail first), not the *destination* (cell-level optimization).

Davies and Lineweaver [4]: Atavistic Theory. See Corollary 4. The atavistic theory correctly identifies the phenotypic and phylostratigraphic pattern. Our framework provides the mechanism it lacks (chain severance) and corrects the causal inference (convergent necessity, not historical reversion). The two theories make different predictions about which genes will be activated in any given cancer: the atavistic theory predicts a coherent ancient program; our framework predicts gene activation determined by $C_n^{(0)}$-level optimization, which overlaps heavily but not completely with the ancient gene set.

Sonnenschein and Soto [15]: TOFT. TOFT is the closest existing framework to our account. It correctly identifies the organizational level (tissue, not cell) at which cancer originates, and correctly argues that proliferation is the default state of cells, with tissue organization as the restraint. Our framework provides TOFT with the formalism it has lacked: the escalation chain is the formal structure of tissue organization, C_n^{eff} is the formal measure of organizational integration, and chain severance is the formal description of organizational breakdown. TOFT's "default proliferative state" is our "necessary action space under cell-level C_n."

Nowell [14]: Clonal Evolution. Clonal evolution describes the dynamics of cancer progression as Darwinian selection among cell populations. Our framework does not replace this account but provides its *attractor*: clonal evolution does not wander randomly through phenotype space but converges on the cell-level optimum determined by $C_n^{(0)}$. The direction of clonal evolution is structurally determined; the stochastic component is in *which cells* reach the attractor first, not *where* the attractor is.

Aktipis et al. [2]: Cancer as Cheating. The evolutionary cooperation framework describes cancer as a breakdown of multicellular cooperation: cells that "cheat" by proliferating at the expense of the collective. Our framework provides the structural basis: cooperation is maintained by the escalation chain (higher organizational levels enforce cooperative behavior by extending C_n^{eff}). "Cheating" is not a strategy adopted by the cell but the necessary outcome of chain severance: a cell that cannot observe tissue-level state cannot cooperate with the tissue, because cooperation requires the knowledge base (K_0) to contain tissue-referencing propositions that are no longer grounded.

Merlo et al. [12]: Cancer as Ecological Process. The ecological framework describes tumor progression as an evolutionary process within the ecosystem of the body. Our framework adds the structural constraint: the "ecosystem" is the escalation chain, and cancer progression is the progressive collapse of that chain. The ecological metaphor is correct but incomplete; it describes dynamics without explaining why those dynamics converge.

8 Testable Predictions

The framework generates predictions that distinguish it from existing theories:

1. **Coupling measurement predicts malignancy.** Quantifying λ_j (the coupling strength between a cell and its tissue context) should predict malignant potential more accurately than mutational burden alone. Cells with low λ_j but few mutations should still progress to malignancy; cells with high mutational burden but intact λ_j should not. This can be tested by measuring intercellular communication (gap junction activity, paracrine signaling responsiveness, contact inhibition) alongside mutational profiling.

2. **Convergent gene expression is constraint-determined, not phylogeny-determined.** Cancer gene expression should converge on a set determined by cell-level optimization, which

mostly but not entirely overlaps with phylogenetically ancient genes. In particular, recently evolved genes that serve cell-level autonomous function (e.g., certain immune evasion mechanisms that evolved in the multicellular era) should be upregulated in cancer despite being phylogenetically recent. The atavistic theory predicts they should not be.

3. **Differentiation therapy should outperform cytotoxic therapy.** Therapies that restore tissue-level coupling (differentiation therapy, microenvironment normalization) should produce more durable responses than therapies that kill cells without restoring coupling, because the former addresses the structural cause (chain severance) while the latter imposes selection pressure on a locally rational optimizer.

4. **Tumor heterogeneity should correlate with severance gradient.** Within a single tumor, phenotypic heterogeneity should correlate with measured coupling strength: cells with residual tissue coupling ($\lambda_j > 0$) should exhibit more differentiated phenotypes than fully decoupled cells. This is testable by spatial transcriptomics combined with intercellular communication assays.

5. **Necessary hallmarks vs. contingent hallmarks.** The framework predicts that some hallmarks are *necessary* under cell-level C_n (proliferation, metabolic autonomy) while others are *contingent* on the specific optimization landscape (angiogenesis induction, specific immune evasion strategies). Necessary hallmarks should appear in every cancer; contingent hallmarks should vary by context. This is testable by systematic cross-cancer hallmark prevalence analysis.

9 Classification of Therapies by Chain Restoration

The framework yields a natural classification of cancer therapies: those that attempt to *restore* the escalation chain (addressing the structural cause) versus those that impose *selection pressure* on a decoupled optimizer (treating the symptom). This classification predicts durability of response.

9.1 Chain-Restoring Therapies

These therapies target the coupling coefficients λ_j, attempting to re-establish the cell's connection to higher organizational levels.

Differentiation therapy (Level $0 \to 1$ restoration). All-trans retinoic acid (ATRA) in acute promyelocytic leukemia (APL) is the paradigm case: it restores the cell's tissue-level identity, re-engaging the differentiation program that chain severance disabled. ATRA does not kill the cancer cells; it restores their coupling to the tissue-level escalation chain, returning tissue-level actions (differentiation, growth arrest) to the accessible action space A_0. The result is a $> 90\%$ cure rate, the highest of any cancer therapy. On our account, this is not a coincidence but a *prediction*: therapies that restore the chain should outperform therapies that do not.

Immune checkpoint inhibition (Level $2 \to 3$ restoration). PD-1/PD-L1 and CTLA-4 inhibitors restore the organism-level surveillance coupling $\lambda_3^{\text{immune}}$. Cancer cells that have downregulated immune recognition markers (severing the organism-level coupling channel) are re-exposed to immune surveillance. The therapy does not kill cancer cells directly; it restores the organizational relationship between the immune system and the tumor. The durable responses observed in a

subset of patients (complete responses lasting years) are consistent with chain restoration: once organism-level surveillance is re-established, it is self-maintaining.

Microenvironment normalization (Level $0 \to 1$ and $1 \to 2$ restoration). Strategies that restore the tissue microenvironment – ECM remodeling, stromal normalization, basement membrane repair – address the tissue-level coupling infrastructure. Low-dose anti-angiogenic therapy (vascular normalization rather than vascular destruction) restores the tissue-organ coupling channel $\lambda_2^{\text{vascular}}$ by normalizing tumor vasculature, improving drug delivery and re-establishing metabolic coupling to the organ level.

Gap junction restoration (Level $0 \to 1$ restoration). Experimental approaches to restore connexin expression (particularly Cx43) directly target the highest-bandwidth cell-tissue coupling channel [1]. The framework predicts that restoring GJIC should suppress malignant behavior even without addressing mutations, because the coupling restoration returns tissue-level actions to the accessible action space.

9.2 Non-Chain-Restoring Therapies

These therapies impose selection pressure on a cell that is already optimizing under cell-level bounded context. The cell's $\gamma > 0$ and $\eta_M > 0$ [11] guarantee that it will adapt.

Cytotoxic chemotherapy. Kills dividing cells without restoring any coupling channel. The surviving subpopulation has undergone selection for resistance – slow-cycling cells, drug efflux pumps, DNA repair enhancement – all of which are within the cell-level action space A_0. Resistance is not a failure of the drug; it is the predictable outcome of imposing selection pressure on a locally rational optimizer.

Radiation therapy. Induces DNA damage. DNA repair is a cell-level action already in A_0; it requires no tissue-level coupling. The cell's gradient descent on U will optimize repair pathways in response to radiation pressure.

Most targeted therapies. Block a specific signaling pathway (e.g., EGFR inhibitors, BRAF inhibitors). The cell, performing gradient descent on U, discovers alternative pathways that restore the blocked function. Pathway redundancy is a feature of cell-level action spaces that have been optimized over billions of years of unicellular evolution. Single-pathway targeting is structurally insufficient because the cell-level action space contains multiple routes to the same optimization target.

Surgery. Removes the decoupled entity entirely. This is effective for localized disease but does not prevent new severance events in remaining tissue, and cannot address disseminated disease (metastasis).

9.3 The Hybrid Strategy

The framework predicts that the optimal therapeutic strategy combines chain restoration with exploitation of structural constraints:

1. Restore coupling where possible (differentiation therapy, immunotherapy, microenvironment normalization).

2. For cells where coupling cannot be restored, target actions that are *necessary* under cell-level $C_n^{(0)}$, actions the cell cannot optimize away from because they are structurally required

by the bounded-context constraint. The Warburg effect (aerobic glycolysis) is a candidate: if glycolytic metabolism is *necessary* under cell-level bounded context (because oxidative phosphorylation requires tissue-level oxygen delivery infrastructure), then targeting glycolysis imposes a constraint the cell cannot evolve around.

3. Use non-chain-restoring therapies (chemotherapy, radiation) only as cytoreductive measures to reduce tumor burden before chain-restoring interventions, not as standalone strategies.

Clinical evidence. The relative success of combination immunotherapy (chain restoration at level $2 \to 3$) over single-agent chemotherapy (non-restoring) in melanoma, non-small cell lung cancer, and renal cell carcinoma is consistent with this prediction. The curative success of ATRA in APL (chain restoration at level $0 \to 1$) versus the near-universal resistance to chemotherapy-only regimens in other acute leukemias provides the strongest existing evidence.

10 Implications

Cancer is not a disease of cells. It is a disease of organizational coupling. The cell itself is functioning correctly; it is optimizing under its bounded context, exactly as the theorems predict any entity must. The pathology is in the severed escalation chain, not in the cell.

The "war on cancer" metaphor is structurally wrong. Cancer cells are not invaders to be destroyed; they are former cooperators who have lost the organizational infrastructure that made cooperation possible. Killing them (chemotherapy, radiation) treats the symptom. Restoring the organizational infrastructure (differentiation therapy, microenvironment normalization, coupling restoration) treats the cause. The distinction is between fighting an entity and repairing a relationship.

Prevention over cure. If cancer is chain severance, then prevention is coupling maintenance. Factors that maintain tissue organization (anti-inflammatory interventions, microenvironment homeostasis, gap junction maintenance) should reduce cancer risk more effectively than factors that reduce mutational burden, though the two are related, since mutations are one mechanism of severance.

Peto's paradox. Large, long-lived organisms do not develop cancer at rates proportional to their cell count and lifespan [13], a puzzle under the somatic mutation theory, which predicts that more cell divisions should produce more mutations and therefore more cancer. The escalation chain framework offers a structural explanation: larger organisms have more elaborate tissue organization, stronger intercellular coupling mechanisms, and more escalation levels. Their C_n^{eff} is higher, meaning more severance is required before cell-level optimization dominates. The additional organizational infrastructure acts as redundancy in the coupling channels: multiple mechanisms must fail before the chain severs completely. Peto's paradox, on this account, is not paradoxical: more cells but also more coupling yields a roughly constant severance rate.

The seven-year puzzle. The clinical observation that many cancers can persist at low grade for years before rapid progression is explained by the severance gradient: partial decoupling ($0 < \lambda_j < 1$) produces a cell that is partially constrained by tissue-level signaling and partially operating under cell-level optimization. This intermediate state can be stable while λ_j remains above a threshold. Progression occurs when λ_j crosses the threshold and cascading decoupling begins. Each loss of coupling removes a constraint that was maintaining the remaining coupling channels.

11 Conclusion

Cancer is not random. Cancer is not ancient. Cancer is what happens when a cell loses its escalation chain.

This paper has applied the bounded-context framework of Papers 1–4 to cancer biology as a speculative but structurally grounded proposal. The framework offers what the three major cancer theories have separately lacked: a formal account of *why* organizational breakdown would produce the specific, convergent phenotype that it does. SMT identifies the mechanisms of chain severance (mutations). The atavistic theory identifies the phenotypic pattern (convergence on unicellular-like behavior). TOFT identifies the organizational level (tissue, not cell). Our framework unifies them: mutations sever the escalation chain (SMT), producing a cell that optimizes under cell-level bounded context (TOFT), which converges on the same action space that shaped unicellular life (atavistic pattern), not by reversion but by structural necessity.

The therapeutic predictions align with where clinical oncology is already heading: differentiation therapy, immunotherapy, and microenvironment normalization are gaining ground over purely cytotoxic approaches. The framework provides a structural reason for this trend: these therapies restore the escalation chain, while cytotoxic therapies impose selection pressure on a locally rational optimizer. The framework generates testable predictions (Section 8) that distinguish this account from existing theories.

The framework's contribution is not empirical but structural: it offers a target for therapy that is not the cell but the coupling, not the action but the context that determines which actions are selected. Whether this structural reasoning maps faithfully onto cancer biology is an empirical question. The predictions are offered in that spirit.

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References

[1] T. Aasen, M. Mesnil, C. C. Naus, P. D. Lampe, and D. W. Laird. Gap junctions and cancer: Communicating for 50 years. *Nature Reviews Cancer*, 16(12):775–788, 2016.

[2] C. A. Aktipis, A. M. Boddy, G. Jansen, U. Hibner, C. C. Maley, and G. S. Wilkinson. Cancer across the tree of life: Cooperation and cheating in multicellularity. *Philosophical Transactions of the Royal Society B*, 370(1673):20140219, 2015.

[3] E. A. Engels, R. M. Pfeiffer, J. F. Fraumeni, B. L. Kasiske, A. K. Israni, J. J. Snyder, R. A. Wolfe, N. P. Goodrich, A. I. Bayakly, G. L. Clarke, and others. Spectrum of cancer risk among US solid organ transplant recipients. *JAMA*, 306(17):1891–1901, 2011.

[4] P. C. W. Davies and C. H. Lineweaver. Cancer tumors as Metazoa 1.0: Tapping genes of ancient ancestors. *Physical Biology*, 8(1):015001, 2011.

[5] T. Kawai, A. B. Cosimi, T. R. Spitzer, N. Tolkoff-Rubin, M. Suthanthiran, S. L. Saidman, J. Shaffer, F. I. Preffer, R. Ber, G. Pascual, and D. H. Sachs. HLA-mismatched renal

transplantation without maintenance immunosuppression. *New England Journal of Medicine*, 358(4):353–361, 2008.

[6] D. Hanahan and R. A. Weinberg. The hallmarks of cancer. *Cell*, 100(1):57–70, 2000.

[7] D. Hanahan and R. A. Weinberg. Hallmarks of cancer: The next generation. *Cell*, 144(5):646–674, 2011.

[8] P. D. McCarthy. Graph structure is necessary for information preservation under bounded context. Manuscript, 2026.

[9] P. D. McCarthy. Convergence of control architectures to escalation under bounded context. Manuscript, 2026.

[10] P. D. McCarthy. Knowledge and action are inseparable under survival pressure. Manuscript, 2026.

[11] P. D. McCarthy. Evaluation-driven descent, encoding permanence, and the structural invariants of intelligence. Manuscript, 2026.

[12] L. M. F. Merlo, J. W. Pepper, B. J. Reid, and C. C. Maley. Cancer as an evolutionary and ecological process. *Nature Reviews Cancer*, 6(12):924–935, 2006.

[13] R. Peto, F. J. C. Roe, P. N. Lee, L. Levy, and J. Clack. Cancer and ageing in mice and men. *British Journal of Cancer*, 32(4):411–426, 1975.

[14] P. C. Nowell. The clonal evolution of tumor cell populations. *Science*, 194(4260):23–28, 1976.

[15] C. Sonnenschein and A. M. Soto. Somatic mutation theory of carcinogenesis: Why it should be dropped and replaced. *Molecular Carcinogenesis*, 29(4):205–211, 2000.

[16] A. S. Trigos, R. B. Pearson, A. T. Papenfuss, and D. L. Goode. Altered interactions between unicellular and multicellular genes drive hallmarks of transformation in a diverse range of solid tumors. *Proceedings of the National Academy of Sciences*, 114(24):6406–6411, 2017.

[17] B. Vogelstein, N. Papadopoulos, V. E. Velculescu, S. Zhou, L. A. Diaz, and K. W. Kinzler. Cancer genome landscapes. *Science*, 339(6127):1546–1558, 2013.

[18] O. Warburg. On the origin of cancer cells. *Science*, 123(3191):309–314, 1956.